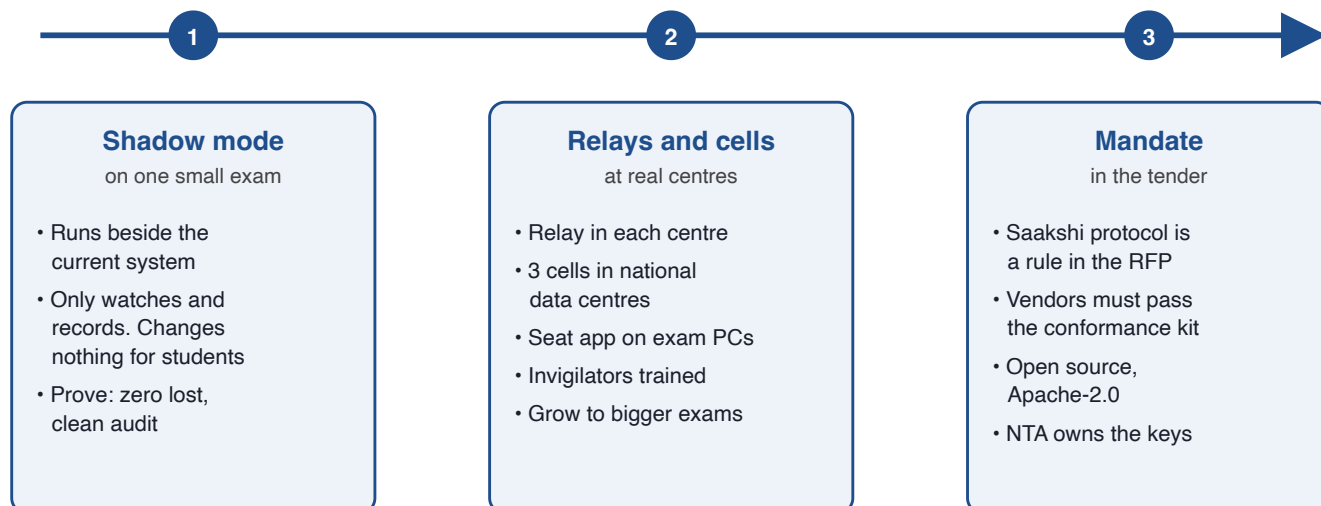


Can it really be built?

Saakshi, implementation and feasibility plan. Short answer: the core already runs. The rest is rollout.

Rollout in three steps



Why shadow mode first? Nobody has to trust us on day one. The old system keeps running. We only compare.

Who does what

Who	Policy and keys	Run control and cells	Run centres	Watch from outside
NTA	Owns	Approves	Approves	Gets reports
NIC	Advises	Owns	-	-
Vendors	-	-	Own, must pass the kit	-
Witnesses	-	-	-	Own, outside the exam path

What already works

Measured on one Apple M5 Mac with 3 real cells and 99 simulated centres. Tested on macOS and Windows.

20,000 candidates in the load test	227,882 entries stored, 0 rejected	20 / 20 chaos runs passed	0 answers lost out of 16,000
44 ms median ingest (p50)	168 ms p99 ingest	~1 s recovery, median 0.96 to 1.23 s	1,000 / 1,000 tampered records located

Checklist

- ✓ Seat app signs and saves every answer, even offline
- ✓ Relay in the centre cannot read answers
- ✓ Three cells verify and countersign; a wiped cell rebuilds from relays
- ✓ Spare relay takes over during an exam
- ✓ Paper released only when 2 of 3 custodians agree; offline unlock code works
- ✓ Candidate moves to another seat and keeps the time they had
- ✓ Audit page shows exactly which record was edited
- ✓ Analytics flags likely cheating; no honest candidate flagged (synthetic data)
- ✓ 476 workspace tests and 76 analytics tests pass

Flow we proved



Honest limits. Everything ran on one machine over loopback, not a real WAN. Analytics numbers use synthetic data. Power-loss and 20k over a real network are not yet measured.

What it needs, and what could go wrong

Hardware we know about

Part	What is needed	Evidence
Exam PC (seat)	Windows 10 x64 or macOS 12, 4 GB RAM, dual core. Camera optional	Minimum spec in plan
Seat load	2.3% CPU median (max 4.3%), 166 MiB memory	Measured
Centre link	About 1.06 MB per candidate per hour	Measured
Centre relay	One small machine per centre, plus a spare running the same program	By design
Cells	3 independent processes in national data centres	Ran 3 on one Mac
Bigger exams	About 10 cells for 1 million candidates at once	Assumption, not measured

Cost. Saakshi is open source (Apache-2.0), so there is no licence fee. A full rupee cost per candidate has not been worked out yet. We will not quote a number we have not measured.

Risks and fixes

Risk	What we do
Camera check fails on some PCs	Invigilator confirms identity by hand instead
Windows hardware arrives late	Packaged app is tested automatically on Windows; demo falls back to the Mac
Rules change and break old records	Protocol v1 is frozen and checked with fixed test vectors
Speed claims do not hold up	Publish only measured numbers; say where they were measured
Fear of false cheating flags	Thresholds published before the exam; a human signs every decision
Too much scope	Shadow mode first; grow only after each step passes
Real network is slower than our test	Seats keep working offline and sync later; early warning when sync lags
One signature does not prove who chose an answer	Stated openly; identity check at entry and invigilator attestation